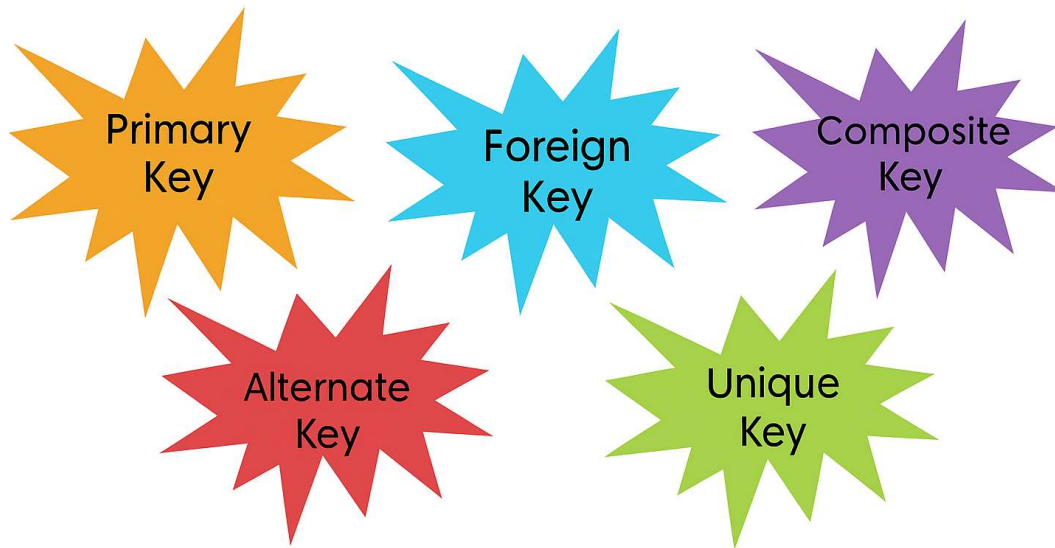


KEYS

◆ What is Key:-

- A key is a single or combination of multiple fields in a table.
- Keys are also used to create a relationship among different database tables.

Types of SQL Keys



◆ Primary Key:-

- **PK** – that uniquely identifies each row in the table.
- Primary key always has a unique data.
- A primary key cannot have null value.
- Syntax:

Create table table_name(column_name datatype(size),
PRIMARY KEY(column_name));

Eg. **create table** student(id **int**, name **varchar**(20),
PRIMARY KEY(id));

◆ Foreign Key:-

- Establish a **link between two tables**.
- Foreign key in one table used to point primary key in another table.
- Syntax:-

```
create table second_table_name(column_name  
datatype(size), PRIMARY KEY(column_name),  
FOREIGN KEY(column_name) REFERENCES  
first_table_name(first_table_column_name));
```

Eg. **create table** student2 (s_id int, **primary key**(s_id),
foreign key(s_id) **references** student1(id));

◆ Composite Key:-

- A primary key that is made by the **combination of more than one attribute** is known as a composite key.
- Syntax:-

```
create table table_name(column_name1 datatype(size),  
column_name2 datatype(size), PRIMARY  
KEY(column_name1, column_name2));
```

Eg. **create table** student(id int, name varchar(20),
PRIMARY KEY(id,name));

◆ Alternate Key:-

- Eg. Id, Roll_no, Email are qualified to become a primary key. But since Id is the primary key, Roll No, Email becomes the alternative key.

Id	Roll_no	Email
101	1	abc@gmail.com

◆ Unique Key:-

- Set of one or more than one fielda/columns of a table that uniquely identify a record in a database table.
- It is little like primary key but it can accept only null value.
- It cannot have duplicate values.
- There may be many unique key constraints for one table, but only one PRIMARY KEY constraint for one table.
- Syntax:-

Create table table_name(column_name1 **datatype**(size),
Column_name2 **datatype**(size),
UNIQUE(column_name1));

Eg. **create table** student(id **int NOT NULL**, name
varchar(20), **UNIQUE**(id));

Difference between Primary and Foreign Key

Primary Key	Foreign Key
➤ Primary key cannot be null.	➤ Foreign key can be null.
➤ Primary key is always unique.	➤ Foreign key can be duplicate.
➤ Primary key uniquely identify a record in the table.	➤ Foreign key is a fields in a table that is primary key in another table.
➤ There is only one primary key in the table.	➤ We can have more than one foreign key in the table.